

ENERGY EFFICIENT COOK STOVE IMPLEMENTATION IN NEPAL

Document Prepared By

Enen Green Services Private Limited

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1 PROJECT DETAILS

1.1 Summary Description of the Project and its Implementation Status

The project is a group project which involves promotion of improved cooking stoves (ICS) to the people of socially deprived community in Nepal. The project intends to avail the households with clean cooking solutions; thereby, displacing the less efficient traditional cooking stoves from the kitchen with stoves of better efficiency. Replacement of the traditional cooking stoves with ICS will reduce the exposure of the family members, specifically women, to the indoor air pollution and therefore result in saving of health related expenses.

All project activity instances included in the group project are located within the country of Nepal only. The estimated annual emission reductions over a 10 year period of first crediting period from initial 18 project activity instances are 59,714 tCO₂e/year. The total emission reductions for 10 years crediting period will be 5,97,140 tCO₂e. All instances included within this group project activity will utilise one single technology of improved cook stove.

As per the calculation in the section, each ICS will reduce emissions by approximately 1.44 tonnes of CO₂e per year (net of leakage and continued use of the baseline stoves). To achieve the initial targeted volume of emission reductions, the project proponent aims to disseminate 41468 ICS in the first phase of this grouped project.

All 18 project activity instances are commissioned. More project activity instances will be included in the grouped project activity in future. These new instances will also be improved cook stoves. These new project activity instances will be of small scale project activity.

Scenario existing prior to the implementation of project activity: The scenario existing prior to the implementation of the project activity instances is the usage of traditional biomass and inefficient cook stoves with poor ventilation, causing excessive indoor air pollution (IAP) and has been a serious health risk for women and children who spend several hours in the kitchen.

1.2 Sectoral Scope and Project Type

The project activity under consideration is a group project activity. The following condition applied to each instance of the group project activity.

Sectoral Scope: 03 Energy Demand.

Project Type: Type II Energy Efficiency Improved Projects

Grouped Project: Yes

1.3 Project Proponent

Organization name	Enen Green services Private Limited
Contact person	Ruchika Sharma
Title	Director
Address	D- 30 A, FF, Ashoka Enclave, Sector 35, Faridabad, Haryana.
Telephone	09953935506

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1.4 Other Entities Involved in the Project

Organization name	Greenway Grameen Infra Pvt. Ltd
Role in the project	Project Participant
Contact person	Neha Juneja
Title	Co-founder & CEO
Address	805, Lodha Supremus -2, Senapati Bapat Marg, Lower parel , Mumbai
Telephone	09953935506
Email	enengreen@gmail.com

1.5 Project Start Date

The first project activity instance of the group project under consideration was commissioned on 01/07/2017. Thus the project start date is 01/07/2017.

1.6 Project Crediting Period

Project Crediting Period: First

Start Date: 01/07/2017(First operating day of the project activity instance ICS).

End date: 01/07/2027

Total Number of Years: 10

1.7 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	x
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
2017	59,714
2018	59,714
2019	59,714

2020	59,714
2021	59,714
2022	59,714
2023	59,714
2024	59,714
2025	59,714
2026	59,714
Total estimated ERs	59,7140
Total number of crediting years	10
Average annual ERs	59,714

1.8 Description of the Project Activity

The project will be broadly implemented in the Nepal. The goal of the group project activity is to disseminate improved cookstoves (ICS) to households in Nepal. Starting with pilot implementation in 2017, it is envisaged to disseminate ICS to households within seven chosen Provinces. There are seven provinces and 77 districts in Nepal. Districts in Nepal are second level of administrative divisions after provinces. Districts are subdivided in municipalities and rural municipalities. The Households in this region of the Nepal are characterized by the consumption of the solid biomass based fuel. More than 62% of the households in these districts use firewood for cooking purpose. Similarly, another 29% use dung cake to meet their daily cooking energy requirements.

The majority of communities are rural and some lack direct road access. This combination of actors drives the high reliance on fuel wood as the primary source of cooking fuel. Despite high prevalence of biogas throughout Nepal, the average penetration rates of biogas and LPG throughout the project boundary is less than 2% each and access to kerosene is also very limited mostly due to high fuel prices.

Project aims to reach disadvantaged people with the clean cooking solutions in form of ICS. Project implementer shall train people appropriately at the local level to cater the households with stove construction. Female stove makers will be trained to the extent possible. Once the installation completes, representatives from the implementing entity will perform paperwork and the stove is recorded in the database. The database will be maintained in MS Excel and contains the following information.

- Name of the User
- Name of the Households Head
- Total Family Members
- District
- Municipality/Rural Municipality

- Ward No.
- Ethnicity
- Installation Date
- Unique ID

The project intends to disseminate a locally made Jumbo and smart ICS to the disadvantaged communities in the project district. The dissemination of ICS will happen phase wise and accordingly the ICS will be included in the project design.

The project will promote Jumbo and smart stove which is an improved stove design. The main design improvement is the use of a prefabricated metallic or ceramic combustion chamber. This combustion chamber ensures consistent quality and durability of the ICS and will improve the lifespan of the stove with consistent performance in terms of efficiency, reduction of indoor air pollution and emissions, and safety. The metal or ceramic combustion chamber is surrounded by an isolative material, then the outer body is constructed in a cube shape, the stove incorporates an L-shaped combustion chamber and pot “skirt” to improve heat transfer and combustion efficiency during cooking activities. The combustion chamber consists of a horizontal fuel magazine and vertical internal chimney. Wood is fed horizontally into the fuel magazine ensuring even combustion from one end and a more easily regulated feed rate. The internal chimney creates draft, accelerating combustions gases from the fire. These gases are then forced through the skirt that surrounds the cook pot. Greater convective heat transfer is the result of improved advection and increased surface area contact. With the high power combustion, emission of indoor air pollutants is minimized rendering a smoke-free kitchen.



1.9 Project



Figure 1: A Greenway Jumbo Stove

Figure 2: A Greenway Smart stove

1.10 Location

All the project activity instances in the proposed grouped project activity would be located within geographical boundaries of Republic of Nepal. Thus geographical area of grouped project is Nepal. The geographical boundary is delineated in the form of extreme geographic coordinates as follows:

Latitude: 28.3949° N

Longitude: 84.12408762° E



1.11 Conditions Prior to Project Initiation

The scenario existing prior to the implementation of the project activity instances is the usage of traditional biomass and inefficient cook stoves with poor ventilation, causing excessive indoor air pollution (IAP) and has been a serious health risk for women and children who spend several hours in the kitchen.

The proposed group project reduces the use and demand for fossil fuels and non-renewable biomass that would have been used in the replaced stove to achieve the same output (i.e cooking daily meals and boiling hot water) with the ICS. This directly leads to reduced GHG emissions. Therefore, the project has been implemented to generated GHG emissions for the purpose of their subsequent reduction, removal or destruction.

1.12 Compliance with Laws, Statutes and Other Regulatory Frameworks

There exists no mandatory law or regulation in the host country requiring the introduction of ICS in Nepal, therefore the project is a voluntary effort by the project proponent.

The project is in compliance with relevant laws, statues and other regulatory frameworks.

The project does not violate any laws, statutes and or other regulatory frameworks in Nepal.

1.13 Ownership and Other Programs

1.13.1 Project Ownership

The improved cook stoves technology installed at site of each of the project activity instances as part of the proposed grouped project activity are under ownership of respective Greenway Gramin Infra Private Limited.

1.13.2 Emissions Trading Programs and Other Binding Limits

The net GHG emission reductions generated by the project activity will not be used for compliance with an emissions trading program or to meet binding limits on GHG emissions.

1.13.3 Other Forms of Environmental Credit

The project proponent hereby corroborates that the project activity has not created or sought or received any other form of environmental credit.

1.13.4 Participation under Other GHG Programs

The project activity by GGIPL has not been registered and is not seeking registration under any other GHG emission program to avail carbon benefits during the crediting period of the project activity

1.13.5 Projects Rejected by Other GHG Programs

The project proponent hereby corroborates that the project activity has not been rejected by any other GHG program

1.14 Additional Information Relevant to the Project

Eligibility Criteria

The project is a grouped project and following are the eligibility criteria for inclusion of new project activity instances into the grouped project activity:

- 1) Applicability Conditions: The project activity instances shall meet applicability conditions for applicable methodology (AMS.II-G version 10) as defined in section 2.2
- 2) Geographical Area: The project activity instances to be included in the grouped project activity will be activities involving implementation of improved cook stoves within Nepal.
- 3) Baseline scenario: All Project Activity Instances shall meet the baseline definition as defined in respective methodology and as explained in section 2.4
- 4) Technology type: The project activity instances to be included in the grouped project activity will be Implementation of Improved cook stoves.
- 5) Additionality: The project activity instances to be added as part of the grouped project will meet additionality criteria as set out in the methodology.

Leakage Management

The project does not involve any transfer of equipments from outside the project boundary to the project boundary. The programme will use the stoves either built on site or manufactured by a

prequalified companies in Nepal. Thus the provision on leakage associated with project boundary is not applicable.

Commercially Sensitive Information

All relevant information for the purpose of Project description is included in VCS PD. All the information disclosed to the validator has been provided in the public version of the report.

Sustainable Development

Project's contribution to Sustainable Development

The contributions of proposed project activity towards sustainable development are explained with indicators viz. social, economical, environmental, technological well-being, legislative and temporal as follows:

Environmental well-being

The project activity will result in the reduction of firewood consumption and emission of green house gases and thus conserve forest and biodiversity.

Social well being

The project activity will pave the way for development and increases the social status and living conditions and the prevailing living standard in the vicinity of the project activity and thus results in empowering the nearby. Also it will contribute to a small increase in the local employment by employing skilled and un-skilled personnel for operation and maintenance of the equipment. The project will reduce the drudgery of women, time saving and the use of saved time for other productive activities.

Economic well being

The project has created a business opportunity during construction phase for local stakeholders such as suppliers, contractors etc. contributing to economic well-being aspects. Further, the project also influences creation of employment opportunities for local people, which would enhance their social status. Sufficiently enhance indoor air quality thereby improving health of women and children and reducing incidences of smoke and fire related injuries and therefore result in saving of health related expenses.

Technological well being

The proposed project activity will promote improved cook stoves that result in reduced fuel consumption and emissions due to cooking and heating water in homes.

Further Information

Legislative:

The Project Proponent has obtained all the relevant approvals required for the establishment and operation of the project activity

Temporal:

EGSPL has not applied for any other kind of renewable or carbon benefit anywhere else for the same.

2 APPLICATION OF METHODOLOGY

2.1 Title and Reference of Methodology

Title: Energy efficiency measures in thermal applications of non-renewable biomass

Type: Type II – Energy Efficiency Improved Projects

Category AMS II.G - Energy efficiency measures in thermal applications of non-renewable biomass /Version 10

Sectoral Scope: 03

Reference: The reference has been taken from the list of the small-scale CDM project activity categories contained in Appendix B of the simplified M&P for small-scale CDM project activities.

<http://cdm.unfccc.int/methodologies/DB/DCO8WRRQVTGLH1GHQBCL035F5M13R8>

The methodology refers to the below CDM tools:

1. Demonstration of additionality of small scale project activities.
2. Calculation of the fraction of non-renewable biomass.

2.2 Applicability of Methodology

The project activity falls under the Type II and Category II G, the applicability conditions in the AMS –II.G. Version 10 is described in the following table.

S. N.	AMS II.G, Version 08 Requirements	Project activity applicability	Means of verification
1.	This methodology comprises efficiency improvements in thermal applications of non-renewable biomass. Examples of applicable technologies and measures include the introduction of high efficiency biomass fired project devices (cookstoves or ovens or dryers) to replace the existing devices and/or energy efficiency improvements in existing biomass fired cookstoves or ovens or dryers.(AMS II G v10, para 2)	This project includes the dissemination of the high efficiency biomass fired cook stoves having the efficiency $\geq 20\%$ compared to traditional stoves. This project will save non-renewable biomass which otherwise would have been consumed by less efficient cooking appliances.	This will be verified by the stove test certificate issued by credited laboratory.
2.	In the case of cookstoves, the methodology is applicable to the introduction of single pot or multi pot portable or in-situ cookstoves with rated efficiency of at least 20 per cent. The Water Boiling Test (WBT) method shall be used to test the efficiency of the cookstove to meet this eligibility requirement, following the requirements indicated in “Data /	This project includes the dissemination of the high efficiency biomass fired cook stoves having the efficiency $\geq 20\%$ compared to traditional stoves. This will be verified by the stove test certificate issued by credited laboratory and submitted to the DOE for validation.	

S. N.	AMS II.G, Version 08 Requirements	Project activity applicability	Means of verification
	Parameter table 11" which details the options for testing and certification as well as supporting documentation (e.g. certificate issued by third party or test results) that needs to be presented to the validating DOE.		
3.	The aggregate energy saving of a single project activity shall not exceed the equivalent of 60 GWh per year or 180 GWh thermal per year in fuel input. (AMS II G v10, para 4)	Annual energy savings from a single ICS is 0.00492 GWh. With this quantity of energy saved the project can include maximum of 36,538 ICS to meet the limit of 60 GW specified by the methodology.	This will be verified using project database.
4.	Project participants are able to show that non-renewable biomass has been used since 31 December 1989, using survey methods or referring to published literature, official reports or statistics. (AMS II G v10, para 5)	The non-renewable biomass has been used in the country since 1989. The information was derived from the field interview with the respondents during the baseline survey.	Default value of fNRB for Nepal.
4.	Leakage related to the non-renewable woody biomass saved by the project activity shall be assessed based on ex post surveys of users and the areas from which this woody biomass is sourced (using 90/30 precision for a selection of samples). The potential source of leakage due to the use/diversion of non-renewable woody biomass saved under the project activity by non-project households/users that previously used renewable energy sources shall be considered. If this leakage assessment quantifies an increase in the use of non-renewable woody biomass by the non-project households/users, that is attributable to the project activity, then $B_{old\ i,j}$ is adjusted to account for the quantified leakage. Alternatively, $B_{y\ saving\ i,j}$ is multiplied by a net to gross adjustment factor of 0.95 to account for leakages, in which case surveys are not required. (AMS II G v10, para 34)	$B_{y\ saving\ i,j}$ will be adjusted by multiplying it with an adjustment factor of 0.95 to account for leakages and in this case surveys will not be required.	Emission reduction calculation spreadsheet.

2.3 Project Boundary



Map of Nepal

Source		Gas	Included?	Justification/Explanation
Baseline	Combustion of non renewable biomass for cooking	CO ₂	Yes	Major source of emissions
		CH ₄	No	Not required by methodology, only CO ₂ emission factor for fossil fuels is considered. Conservative Assumption.
		N ₂ O	No	Not required by methodology, only CO ₂ emission factor for fossil fuels is considered. . Conservative Assumption.
		Other	NA	
Project	Combustion of non renewable biomass for cooking	CO ₂	Yes	Major source of emissions
		CH ₄	No	Not required by methodology, only CO ₂ emission factor for fossil fuels is considered.
		N ₂ O	No	Not required by methodology, only CO ₂ emission factor for fossil fuels is considered.
		Other		

2.4 Baseline Scenario

The actual baseline is the use of non renewable biomass. Methodology AMS II.G states that “in the absence of the project activity, the baseline scenario would be the use of fossil fuels for meeting similar thermal energy needs”. This is a conservative approach to determine the baseline emissions. In absence of the project activities, the intended beneficiaries of the project would continue using the traditional inefficient cooking stoves, consuming high quantity of non-renewable biomass.

2.5 Additionality

The additionality of the project is demonstrated by following the criteria outlined in “Guidelines for demonstrating additionality of small scale project activities” (version 10), Annex 14, EB 83. The paragraph 2(c) of this guideline states: Based on the guidelines, the table below discusses the applicability of the project under discussion as additional.

Specify the methodology or activity requirement or product requirement that establish deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).	Guidelines for demonstrating additionality of small scale project activities” (version 10), Annex 14, EB 83. Paragraph 11 (c) defines additionality if “Project activities solely composed of isolated units where the users of the technology/measure are households or communities or Small and Medium Enterprises (SMEs) and where the size of each unit is no larger than 5% of the small-scale CDM thresholds.”
Describe how the proposed project meets the criteria for deemed additionality.	▪ The annual energy saving per unit (ICS) is 0.004 GWh thermal which is less than 9 MWhth (i.e. less than 5% of the small scale CDM threshold) ▪ Project is considered additional provided that the project is limited to 40 Giga-watt hours of savings per year, which is equivalent to 120 Giga-watt hours thermal per year. Thus, because the project cannot exceed 180 Giga-watt hour thermal energy savings per year, as stated in the eligibility criteria, the project is considered additional.

2.6 Methodology Deviations

No deviations from the applied approved methodology have been taken for the project activity.

3 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

3.1 Baseline Emissions

In accordance with AMS.II-G version 10 paragraph 16, it is assumed that in the absence of the project activity, the baseline scenario would be the use of fossil fuels for meeting similar thermal energy needs. The actual baseline scenario is the use of NRB. Since NRB has higher carbon intensity than the fossil fuels proposed in AMS II.G, this assumption reduces the emission reductions significantly, making the outcome more conservative. The detailed calculations are provided in section 5.2 of the VCS PD

3.2 Project Emissions

There is no project emissions considered.

3.3 Leakage

As per the paragraph 30 of the methodology AMS II.G/v010, the following leakages are to be considered:

The use/diversion of non-renewable woody biomass saved under the project activity by non-project households/users that previously used renewable energy sources. If this leakage assessment quantifies an increase in the use of non-renewable woody biomass used by the non-project households/users that is attributable to the project activity, then Bold is adjusted to account for the quantified leakage. Alternatively, B_{old} is multiplied by a net to gross adjustment factor of 0.95 to account for leakages, in which case surveys are not required. PP has opted to use the adjustment factor of 0.95 in order to account for leakage.

As per the paragraph 32 of the methodology AMS II.G/v10, if equipment currently being utilised is transferred from outside the boundary to the project activity, leakage is to be considered.

The project does not involve any transfer of equipments from outside the project boundary to the project boundary. The programme will use the stoves either built on site or manufactured by a prequalified companies in Nepal. Thus the provision on leakage associated with project boundary is not applicable

3.4 Estimated Net GHG Emission Reductions and Removals

According to paragraph 15 of methodology AMS II.G/v10, emission reductions would be calculated as:

$$ER_y = \sum_i ER_{y,i} \quad (1)$$

Where:

i = Indices for the situation where more than one type of project device is introduced to replace the pre-project devices
 ER_y = Emission reductions during year y in tCO₂e
 $ER_{y,i}$ = Emission reductions by project device of type i during year y in tCO₂e

For household cook stoves (as guided by the methodology):

$$ER_{y,i} = \sum_{a=i}^{a=y} B_{y,savings,i,a} \times N_{y,i} \times \mu_y \times f_{NRB,y} \times NCV_{biomass} \times EF_{projected_fossil\ fuel} \quad (2)$$

Where:

A : 'a' is the indices for age (in years) of the cook stoves for the age (in years) of the cook stoves that are operating in the year 'y' of the crediting period
 $B_{y,savings,i,a}$: Quantity of woody biomass that is saved in tonnes per cook stove device of type i and age a in year y
 $f_{NRB,y}$: Fraction of woody biomass saved by the project activity in year y that can be established as non-renewable biomass using survey methods or government data or default country specific fraction of non renewable woody biomass (f_{NRB}) values available on CDM website¹

¹Default values endorsed by designated national authorities and approved by the Board are available at <<http://cdm.unfccc.int/DNA/fNRB/index.html>>.

$NCV_{biomass}$	:	Net calorific value of the non-renewable woody biomass that is substituted (IPCC default for wood fuel, 0.015 TJ/tonne, wet basis)
$EF_{projected_fossilfuel}$	:	Emission factor for the substitution of non-renewable woody biomass by similar consumers. Use a value of 81.6 tCO ₂ /TJ ²
$N_{y,i,a}$	:	Number of project devices of type i and age a operating in year y
$\mu_{y,i}$	:	Number of days of utilization of the project device during the year 'y'.

As stipulated in the paragraph 16 of AMS II.G/v10, By,savings can be determined using one of the following options:

Option 1:

$$B_{y,saving} = B_{old,i} - B_{a=1,i,KPT} \times \Delta B_{y,i,a} \quad (3)$$

Where:

$B_{old,i}$	:	Annual quantity of woody biomass used in the absence of the project activity to generate thermal energy equivalent to that provided by the project device of type i, if the project device operate throughout the year y. Value in tonnes per year per device of type i.
$B_{a=1,i,KPT}$	:	Annual quantity of woody biomass used in tonnes per device of type i, measured as per the KPT protocol, for the initial efficiency determined in the year of its installation (a=1). The KPT shall be carried out in accordance with national standards (if available) or international standards or guidelines (e.g. the KPT procedures specified by the Partnership for Clean Indoor Air (PCIA) < http://www.pciaonline.org/node/1049 >)

Option 2³:

$$B_{y,savings,i,a} = B_{old} \times \left(1 - \frac{\eta_{old}}{\eta_{new,i,a=1} \times \Delta \eta_{y,i,a}} \right) \quad (4)$$

$$B_{y,savings,i,a} = B_{old} \times \left(\frac{\eta_{new,i,a=1} \times \Delta \eta_{y,i,a}}{\eta_{old}} - 1 \right) \quad (5)$$

Where:

B_{old}	:	Quantity of woody biomass used in the absence of the project activity in tonnes per device
$B_{y=1,new,i,survey}$	:	Annual quantity of woody biomass used by project devices in tonnes per device of type i, determined in the first year of introduction of the devices (e.g. during the first year of the crediting period, y=1) through a sample survey. Sample surveys to estimate this parameter, that are solely based on questionnaires or interviews (i.e. that do not implement measurement campaigns) may only be used if the following

² This value represents the emission factor of the substitution fuels likely to be used by similar users, on a weighted average basis. It is assumed that the mix of present and future fuels used would consist of a solid fossil fuel (lowest in the ladder of fuel choices), a liquid fossil fuel (represents a progression over solid fuel in the ladder of fuel use choices) and a gaseous fuel (represents a progression over liquid fuel in the ladder of fuel use choices). Thus a 50% weight is assigned to coal as the alternative solid fossil fuel (96 tCO₂/TJ) and a 25% weight is assigned to both liquid and gaseous fuels (71.5 tCO₂/TJ for Kerosene and 63.0 tCO₂/TJ for Liquefied Petroleum Gas (LPG)).

³Based on whether $\eta_{new,y}$ or $B_{y,new,survey}$ is used for monitoring, either the Equation (3) or (4) can be used.

conditions are satisfied.

- a) Pre-project devices have been completely decommissioned and only efficient project devices are exclusively used in the project households;
- b) If multiple devices are used in the project, it is possible from the results of the survey questions to clearly differentiate the quantity of woody biomass being used by each device. In other words, if more than one device, or another device that consume woody biomass, are in use in project households, then the sample survey needs to distinguish the quantity of biomass used by the project device and the other devices that use biomass.

η_{old} : Efficiency of the device being replaced (fraction), determined using one of the following options:

1. Measured using representative sampling methods or based on literatures reporting results of measurements relevant for the type of pre-project devices. Use weighted average values (taking the amount of woody biomass consumed by each device as the weighting factor) if more than one type of device is being replaced;
2. A default value of 0.10 may be optionally used if the replaced system is a three stone fire using firewood (not charcoal), or a conventional device with no improved combustion air supply or flue gas ventilation, i.e. without a grate or a chimney; for other types of devices a default value of 0.2 may be optionally used.

$\eta_{new,i,a=1}$: Thermal efficiency of the device of type i being deployed as part of the project activity (fraction), using the Water Boiling Test (WBT) protocol carried out in accordance with national standards (if available) or international standards or guidelines⁴, for the initial efficiency determined in the year of its installation (a=1)

$\Delta\eta_{y,i,a}$: Factor to consider the efficiency loss of the project device type i due to its aging at the year y, as expressed as follows:
 $\Delta\eta_{y,i,a} = \eta_{new,i,a} / \eta_{new,i,a=1}$

where $\eta_{new,i,a}$ is thermal efficiency of device i with age a determined using WBT and $\eta_{new,i,a=1}$ is the thermal efficiency of the device at its first year of operation. $\Delta\eta_{y,i,a}$ may be determined through sample surveys of the project device type i for batches of stoves with the same age at each year of crediting period. Alternatively, the monitoring may determine annually the thermal efficiency of devices installed at the first year of the crediting period, and the efficiency loss of the population of devices later on.

Option 3:

$$B_{y,savings,i,a} = B_{old,i} \times \left(1 - \frac{SC_{new,i,a=1} \times \Delta SC_{y,i,a}}{SC_{old}} \right) \quad (6)$$

Where:

⁴In all cases the testing protocol shall be the same for both the device being replaced and the device being deployed

$SC_{new,i,a=1}$	:	Specific fuel consumption or fuel consumption rate of the devices of type i deployed as part of the project i.e. fuel consumption per quantity of item/s processed (e.g. food cooked) or fuel consumption per hour, respectively, for the initial efficiency determined in the year of its installation (a=1). Specific fuel consumption or fuel consumption rate shall be determined using the same CCT protocol used to test the pre-project devices. If more than one project devices are necessary to replace the pre-project device, woody biomass consumption should be calculated per device (taking the amount of woody biomass consumed by each device as the weighting factor).
SC_{old}	:	Specific fuel consumption or the fuel consumption rate of the pre-project devices, that is fuel consumption per quantity of item/s processed (e.g. food cooked) or fuel consumption per hour respectively. Specific fuel consumption or fuel consumption rate are to be determined using CCT protocol carried out in accordance with national standards (if available) or international standards or guidelines (e.g. the CCT procedures specified by the partnership for clean indoor air (PCIA).
$\Delta SC_{y,i,a}$	:	Factor to consider the efficiency loss of the project device type i due to its aging at the year y, as expressed as follows: $\Delta SC_{y,i,a} = SC_{new,i,a=1} / SC_{new,i,a}$

where, $SC_{new,i,a}$ is the specific fuel consumption of the device i with age a determined using the CCT and $SC_{new,i,a=1}$ is the specific fuel consumption of the device at its first year of operation $\Delta SC_{y,i,a}$ may be determined through sample surveys of project device type I for batches of stoves with the same age at each year of the crediting period. Alternatively, the monitoring may determine annually the specific fuel consumption of the devices installed at the first year of the crediting period, and efficiency loss of this population may be used to correct the initial efficiency of the population of devices installed later on.

The methodology requires choosing one of the three options mentioned above for calculating the "Quantity of woody biomass that is saved (By,savings)". Option 2 equation (4) has been selected. As stipulated in the paragraph 19 of AMS II.G/v06, B_{old} can be determined using one of the following options:

1. Estimated as the average annual consumption of the woody biomass per device (tonnes/year). This can be derived from historical data or a survey of local usage,
OR
2. Calculated from the thermal energy generated in the project activity as:

$$B_{old} = \frac{HG_{p,y}}{NCV_{biomass} * \eta_{old}} \quad (6)$$

Where:

$HG_{p,y}$: Amount of thermal energy generated by the project devices in year y (TJ), if the thermal output of the devices can be directly measured

OR

1. A default value of 0.5 tonnes woody biomass per capita per year may be used. The household occupancy shall be determined based on literature or sample surveys conducted for the target project area as per "Standard for sampling and surveys for CDM project activities and program of activities".

The methodology requires choosing one of the three options mentioned above for calculating the “Quantity of woody biomass used in the absence of the project activity (Bold)”. Since, it is difficult and not cost effective to monitor the thermal energy generated in the project activity, Option 1 has been selected, as it is feasible to determine Bold based on number of devices multiplied by the estimate of average annual consumption of woody biomass per device substituted (tonnes/year), which can be derived from historical data or estimated using survey methods. As the project involves improved cooking stoves, which are owned and operated by households, the term individual device is meant to represent individual improved cooking stove.

Differentiation between NRB and renewable biomass and determining $f_{NRB,y}$

The methodology requires the Project Participants to determine the shares of renewable and non-renewable woody biomass in Bold (the quantity of woody biomass used in the absence of the project activity) the total biomass consumption using nationally approved methods (e.g. surveys or government data if available) and then determine $f_{NRB,y}$ as described below. The following principles shall be taken into account:

Demonstrably renewable woody biomass⁵ (DRB)

According to AMS IIG/v10 para 28, woody⁶ biomass is “renewable” if one of the following two conditions is satisfied:

1. The woody biomass is originating from land areas that are forests⁷ where:
 - (a) The land area remains a forest; and
 - (b) Sustainable management practices are undertaken on these land areas to ensure, in particular, that the level of carbon stocks on these land areas does not systematically decrease over time (carbon stocks may temporarily decrease due to harvesting); and
 - (c) Any national or regional forestry and nature conservation regulations are complied with.
2. The biomass is woody biomass and originates from non-forest areas (e.g., croplands, grasslands) where:
 - (a) The land area remains as non-forest or is reverted to forest; and
 - (b) Sustainable management practices are undertaken on these land areas to ensure in particular that the level of carbon stocks on these land areas does not systematically decrease over time (carbon stocks may temporarily decrease due to harvesting); and
 - (c) Any national or regional forestry, agriculture and nature conservation regulations are complied with.

Non-renewable biomass:

According to AMS IIG/v10 para 27, non-renewable woody biomass (*NRB*) is the quantity of woody biomass used in the absence of the project activity (B_{old}) minus the *DRB* component, as long as at least two of the following supporting indicators are shown to exist:

- A trend showing an increase in time spent or distance travelled for gathering fuel-wood, by users (or fuel-wood suppliers) or alternatively, a trend showing an increase in the distance the fuel-wood is transported to the project area;
- Survey results, national or local statistics, studies, maps or other sources of information, such as remote-sensing data, that show that carbon stocks are depleting in the project area;
- Increasing trends in fuel wood prices indicating a scarcity of fuel-wood;

⁵ This definition uses elements of annex 18, EB 23.

⁶ In cases of charcoal produced from woody biomass, the demonstration of renewability shall be done for the areas where the woody biomass is sourced.

⁷ The forest definitions as established by the country in accordance with the decisions 11/CP.7 and 19/CP.9 should apply.

- Trends in the types of cooking fuel collected by users that indicate a scarcity of woody biomass.

Thus the fraction of woody biomass saved by the project activity in year y that can be established as non-renewable is:

$$f_{NRB,y} = \frac{NRB}{NRB + DRB} \quad (1)$$

As per EB 67, Annex 22 "Information note Default values of fraction of non-renewable biomass for Least Developed Countries and Small Island Developing States (version 01.0)", a default value of f_{NRB} 86% has been applied. Therefore the default value of f_{NRB} is 86%

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
2017-2018	59,714	0	0	59,714
2018-2019	59,714	0	0	59,714
2019-2020	59,714	0	0	59,714
2020-2021	59,714	0	0	59,714
2021-2022	59,714	0	0	59,714
2022-2023	59,714	0	0	59,714
2023-2024	59,714	0	0	59,714
2024-2025	59,714	0	0	59,714
2025-2026	59,714	0	0	59,714
2026-2027	59,714	0	0	59,714
Total	59,7140	0	0	59,7140

4 MONITORING

4.1 Data and Parameters Available at Validation

Data / Parameter	B _{old,HH}
Data unit	tonnes/household/year
Description	Quantity of woody biomass used per ICS in the absence of the project activity
Source of data	Baseline Survey
Value applied	2.50

Justification of choice of data or description of measurement methods and procedures applied	Default value baseline firewood consumption i.e. 0.5 tons/capita/year is applied. Baseline Survey shows there are averagely 4.03 persons / household. Based on the above, $B_{old,p} = 0.5 \text{ tons/capita/year} * 4.03 \text{ persons/household} = 2.015 \text{ tons/household/year}$
Purpose of Data	Calculation of Quantity of woody biomass that is saved in tonnes per device
Comments	This parameter shall remain fixed for the monitoring periods.

Data / Parameter	$N_{p,HH}$
Data unit	Number
Description	Average number of persons served per household prior to project implementation
Source of data	Baseline Survey
Value applied	5
Justification of choice of data or description of measurement methods and procedures applied	Scott Wilson Nepal Pvt. Ltd. Report "Preparation of Baseline for Improved Cook Stoves (ICS) Programme in Hilly Districts of Far Western Development Region". September 2012
Purpose of Data	Calculation of Quantity of woody biomass that is saved in tonnes per device
Comments	This parameter shall remain fixed for the monitoring periods.

Data / Parameter	$B_{old,p}$
Data unit	tonnes/person/year
Description	Annual quantity of woody biomass that would have been used per person in the household in the absence of the project activity to generate useful thermal energy equivalent to that provided by the project devices
Source of data	Baseline Survey
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	A default value of 0.5 tonnes/capita per year may be used. This option is limited to household project devices (not eligible for oven and dryers)
Purpose of Data	Calculation of Quantity of woody biomass that is saved in tonnes per device
Comments	This parameter shall remain fixed for the monitoring periods.

Data / Parameter	$f_{NRB,y}$
Data unit	Percentage
Description	Fraction of woody biomass saved by the project activity during year y that can be established as non-renewable biomass
Source of data	"Default values of fraction of non-renewable biomass for Least Developed Countries and Small Island Developing States (version 01.0)" EB 67 Annex 22.
Value applied	86%
Justification of choice of data or description of measurement methods and procedures applied	Any changes/ revisions to this value would be monitored and if new default values are approved by the CDM EB the value of the fraction of NRB used will be updated.
Purpose of Data	Calculation of share of non-renewable biomass
Comments	This parameter shall remain fixed for the monitoring periods.

Data / Parameter	η_{old}
Data unit	Percentage
Description	Efficiency of the system being replaced (Traditional Cooking Stoves)
Source of data	Table 12 of AMS II.G/v08
Value applied	10%
Justification of choice of data or description of measurement methods and procedures applied	The default value of 0.10 is used as the replaced system is a three stone fire, or a conventional device with no improved combustion air supply or flue gas ventilation, i.e. without a grate or a chimney.
Purpose of Data	Calculation of Quantity of woody biomass that is saved in tonnes per device
Comments	This parameter shall remain fixed for the monitoring periods.

Data / Parameter	$NCV_{biomass}$
Data unit	TJ/tonne
Description	Net calorific value of the non-renewable woody biomass that is substituted
Source of data	Table 12 of AMS II.G/v09
Value applied	0.0156
Justification of choice of data or description of	As per the methodology AMS II.G/v09

measurement methods and procedures applied	
Purpose of Data	Emission reduction calculation
Comments	This parameter shall remain fixed for the monitoring periods.

Data / Parameter	$EF_{projected_fossilfuel}$
Data unit	tCO ₂ /TJ
Description	Emission factor for the substitution of non-renewable woody biomass by similar consumers
Source of data	Table 17 of AMS II.G/v09
Value applied	63.0
Justification of choice of data or description of measurement methods and procedures applied	This value represents the emission factor of the substitution fuels likely to be used by similar users, on a weighted average basis. The value is calculated, based on the global average ratio of cooking fuels (the normalized ratio of kerosene and liquefied petroleum gas (LPG) excluding coal), i.e. 9 per cent for kerosene (71.5 t CO ₂ /TJ) and 91 per cent for LPG (63.0 t CO ₂ /TJ).
Purpose of Data	Emission reduction calculations
Comments	This parameter shall remain fixed for the monitoring periods.

Data / Parameter	L_y
Data unit	Fraction
Description	Leakage adjustment factor
Source of data	Paragraph 29 of AMS II.G/v09
Value applied	0.95
Justification of choice of data or description of measurement methods and procedures applied	As per the methodology AMS II.G/v06, B_{old} can be multiplied by a net to gross adjustment factor of 0.95 to account for leakages, in which case surveys are not required.
Purpose of Data	Emission reduction calculations
Comments	This parameter shall remain fixed for the monitoring periods.

4.2 Data and Parameters Monitored

Data / Parameter	$N_{y,i,a}$
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Data unit	Number
Description	Number of project devices of type i and age a that are operating in year y
Source of data	Annual ICS users' survey
Description of measurement methods and procedures to be applied	The information on the number of devices operational is determined by the ICS users' survey by an independent third party through monitoring of the user households drawn at random sample
Frequency of monitoring/recording	Annual
Value applied	41468
Monitoring equipment	Annual user survey
QA/QC procedures to be applied	During the annual users' survey, an independent third party will inspect representative sample households to check if the devices are operating or not. Sample for this survey will be drawn as per the "Guidelines for sampling and surveys for CDM project activities and programme of activities, version 3 (EB 75, annex 8)".
Purpose of data	Emission reduction calculation
Calculation method	
Comments	If households are found continuing the use of traditional stoves, such households will be counted for having non-operational project device and equal proportion of households will not be counted towards emission reductions. Nevertheless, some households may retain their traditional stoves for religious reason where the traditional stoves are not used, the annual ICS users' survey will capture this information and it will be used accordingly while calculating the emission reductions.

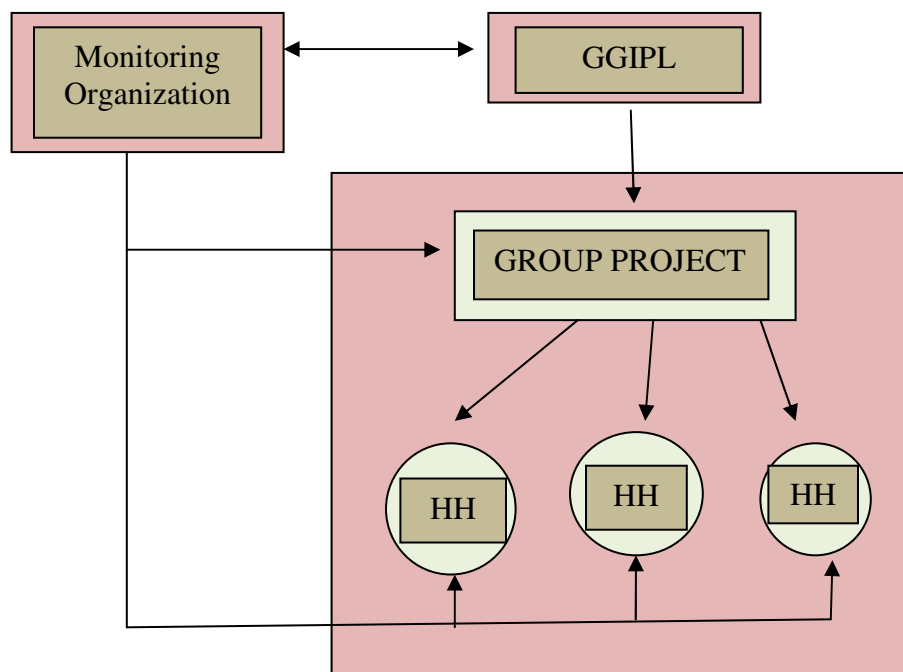
Data / Parameter	$\eta_{\text{new},i,a}$
Data unit	Fraction
Description	Efficiency of the device of type i being deployed as part of the project activity.
Source of data	Water Boiling Test report conducted by an independent third party during the annual ICS users' survey
Description of measurement methods and procedures to be applied	The efficiency test will be carried out for different types and models of ICS. The value obtained from the test will be used to calculate the emission reductions of the systems for that year of operation. Sample for this survey will be drawn as per the "Guidelines for sampling and surveys for CDM project activities

	and programme of activities, version 3 (EB 75, annex 8)".
Frequency of monitoring/recording	Annual
Value applied	31.17%
Monitoring equipment	-
QA/QC procedures to be applied	During the annual users' survey, an independent third party will inspect representative sample households to check if the devices are operating or not. Sample for this survey will be drawn as per the "Guidelines for sampling and surveys for CDM project activities and programme of activities, version 3 (EB 75, annex 8)".
Purpose of data	The monitored value of this parameter will be used in the determination of the ex-post emission reduction.
Calculation method	Third party certification.
Comments	If any sample stove is found to be operating below the 20% efficiency, the proportionate number of stoves will be considered non-operational and not accounted for ER calculation.

4.3 Monitoring Plan

The purpose of the monitoring plan is to ensure successful monitoring of the emission reductions of the proposed project during its crediting period. The overall monitoring will be managed by the project implementer CGIPL, Nepal

Activities and performance related to emissions reduction are monitored by EGSPL. The PP proposed the following structure for data monitoring, collection, data archiving and calibration of equipments for this project activity instance. The team comprise will follow the below organizational structure.



The monitoring activities will involve data collection during distribution as well as usage information post distribution. The data collected during distribution will involve information about the stove, the purchaser and location to enable one to uniquely identify each ICS unit and avoid double counting. This will form what is called the sales record. Greenway will distribute the stoves, as well as carry out the monitoring activities that occur during the distribution of stoves. Greenway or through their associates will ensure that the distributor is trained on how to capture the sales data. The project activity implementer is fully responsible to ensure the correct distribution process and data gathering is followed. In turn, the distributing agent will link remuneration to the complete and accurate collecting of information during the distribution of stoves.

The following information will be recorded at the time of distribution of ICS to the user:

- Name of the User
- Name of the Households Head
- Total Family Members
- District
- Municipality/Rural Municipality
- Ward No.
- Ethnicity
- Installation Date
- Unique ID

However, during post distribution, monitoring activities will involve selecting a sample of stove from the sales record and visiting the premises where these stoves are located to monitor key parameters pertinent in ER Calculation. The individuals carrying out the monitoring activities will follow instructions provided during training, to check and record the following key parameters:

- Check the efficiency of the ICS.
- Check if project stoves are operational and in use
- Check if there is any on-going use of replaced stoves.

Data collecting & handling is conducted in a transparent way to secure high quality of recording and storing of data. Data collected and monitored are stored electronically in a secure and retrievable manner for at least two years after the end of the project crediting period. Uncertainty related to data handling (if any) would be rectified, if necessary by revising monitoring procedures. The changes would be approved by a verifier (e.g. Designated Operational Entity)

4.4 No Net Harm

Since the technology i.e. cook stove under the project is very simple and easy to use and maintain. There is very less chance of the technology failure and equipment damage that may impact the applicability of the methodology.

4.5 Environmental Impact

- Environmental Benefits: Reduction in firewood consumption and emission of greenhouse gases, forest and biodiversity conservation.
- Economic Benefits: Employment creation and saving of health cost.
- Air quality: Sufficiently enhance indoor air quality thereby improving health of women and children and reducing incidences of smoke and fire related injuries.
- Social Benefit : Reduces drudgery of women, time saving and the use of saved time for other productive activities

4.6 Local Stakeholder Consultation

Invitations for stakeholder consultation meeting were sent out by GGIPL via an advertisement in the newspaper and personal letters. A public notice for the same was also affixed at main gate of GGIPL. Stakeholders were identified as those whose activities directly or indirectly impact the project, and those who were to be impacted by the project activities (full lists of attendance are in the Local Stakeholder Consultation Report).

GGIPL organized the stakeholder consultation meeting on 25th December, 2018 .The meeting was attended by the members from the industrial complex, consultants, suppliers, contractors and the local residents⁸ in the area. The objective of the meeting was to inform the stakeholders about the environmental and social impacts of the project activity and discuss their concerns regarding the same, if any. The comments and suggestions were invited throughout the period of Stakeholders meet.

⁸ list of attendees

The overall response to the project, from all invited stakeholders, was encouraging and positive. In all, no adverse reactions, comments, or clarifications were received during the Stakeholder Consultation process. The project participants did not recognize any need to make changes to the project design or monitoring plan.

4.7 Public Comments

The overall response to the project, from all invited stakeholders, was encouraging and positive. In all, no adverse reactions, comments, or clarifications were received during the Stakeholder Consultation process. The project participants did not recognize any need to make changes to the project design or monitoring plan.

5 ACHIEVED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Data and Parameters Monitored

Data / Parameter	$N_{y,i,a}$
Data unit	Number
Description	Number of project devices of type i and age a that are operating in year y
Value applied:	41468
Comments	If households are found continuing the use of traditional stoves, such households will be counted for having non-operational project device and equal proportion of households will not be counted towards emission reductions. Nevertheless, some households may retain their traditional stoves for religious reason where the traditional stoves are not used, the annual ICS users' survey will capture this information and it will be used accordingly while calculating the emission reductions.

Data / Parameter	$\eta_{new,i,a}$
Data unit	Fraction
Description	Efficiency of the device of type i being deployed as part of the project activity.
Value applied:	31.17 %
Comments	If any sample stove is found to be operating below the 20% efficiency, the proportionate number of stoves will be considered non-operational and not accounted for ER calculation.

5.2 Baseline Emissions

In accordance with AMS.II-G version 10 paragraph 16, it is assumed that in the absence of the project activity, the baseline scenario would be the use of fossil fuels for meeting similar thermal energy needs. The actual baseline scenario is the use of NRB. Since NRB has higher carbon intensity than the fossil fuels proposed in AMS II.G, this assumption reduces the emission reductions significantly, making the outcome more conservative. The detailed calculations are provided in section 5.3 of the VCS PD

5.3 Project Emissions

There is no project emissions considered.

5.4 Leakage

As per the paragraph 30 of the methodology AMS II.G/v010, the following leakages are to be considered:

The use/diversion of non-renewable woody biomass saved under the project activity by non-project households/users that previously used renewable energy sources. If this leakage assessment quantifies an increase in the use of non-renewable woody biomass used by the non-project households/users that is attributable to the project activity, then B_{old} is adjusted to account for the quantified leakage. Alternatively, B_{old} is multiplied by a net to gross adjustment factor of 0.95 to account for leakages, in which case surveys are not required. PP has opted to use the adjustment factor of 0.95 in order to account for leakage.

As per the paragraph 32 of the methodology AMS II.G/v10, if equipment currently being utilised is transferred from outside the boundary to the project activity, leakage is to be considered.

The project does not involve any transfer of equipments from outside the project boundary to the project boundary. The programme will use the stoves either built on site or manufactured by a prequalified companies in Nepal. Thus the provision on leakage associated with project boundary is not applicable

5.5 Net GHG Emission Reductions and Removals

According to paragraph 15 of methodology AMS II.G/v10, emission reductions would be calculated as:

$$ER_y = \sum_i ER_{y,i} \quad (1)$$

Where:

i = Indices for the situation where more than one type of project device is introduced to replace the pre-project devices
 ER_y = Emission reductions during year y in tCO₂e
 $ER_{y,i}$ = Emission reductions by project device of type i during year y in tCO₂e

For household cook stoves (as guided by the methodology):

$$ER_{y,i} = \sum_{a=1}^{a=y} B_{y,savings,i,a} \times N_{y,i} \times \mu_y \times f_{NRB,y} \times NCV_{biomass} \times EF_{projected_fossil_fuel} \quad (2)$$

Where:

A	:	'a' is the indices for age (in years) of the cook stoves for the age (in years) of the cook stoves that are operating in the year 'y' of the crediting period
$B_{y,savings,i,a}$	:	Quantity of woody biomass that is saved in tonnes per cook stove device of type i and age a in year y
$f_{NRB,y}$	:	Fraction of woody biomass saved by the project activity in year y that can be established as non-renewable biomass using survey methods or government data or default country specific fraction of non renewable woody biomass (f_{NRB}) values available on CDM website ⁹
$NCV_{biomass}$	:	Net calorific value of the non-renewable woody biomass that is substituted (IPCC default for wood fuel, 0.015 TJ/tonne, wet basis)
$EF_{projected_fossilfuel}$	:	Emission factor for the substitution of non-renewable woody biomass by similar consumers. Use a value of 81.6 tCO ₂ /TJ ¹⁰
$N_{y,i,a}$	:	Number of project devices of type i and age a operating in year y
$\mu_{y,i}$	:	Number of days of utilization of the project device during the year 'y'.

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
2017-2018	59,714	0	0	59,714
2018-2019	59,714	0	0	59,714
2019-2020		0		
2020-2021		0		
2021-2022		0		
2022-2023		0		
2023-2024		0		
2024-2025		0		
2025-2026		0		

⁹Default values endorsed by designated national authorities and approved by the Board are available at <<http://cdm.unfccc.int/DNA/fNRB/index.html>>.

¹⁰ This value represents the emission factor of the substitution fuels likely to be used by similar users, on a weighted average basis. It is assumed that the mix of present and future fuels used would consist of a solid fossil fuel (lowest in the ladder of fuel choices), a liquid fossil fuel (represents a progression over solid fuel in the ladder of fuel use choices) and a gaseous fuel (represents a progression over liquid fuel in the ladder of fuel use choices). Thus a 50% weight is assigned to coal as the alternative solid fossil fuel (96 tCO₂/TJ) and a 25% weight is assigned to both liquid and gaseous fuels (71.5 tCO₂/TJ for Kerosene and 63.0 tCO₂/TJ for Liquefied Petroleum Gas (LPG)).

2026-2027		0		
Total	1,19,428	0		1,19,428

APPENDIX X: <TITLE OF APPENDIX>